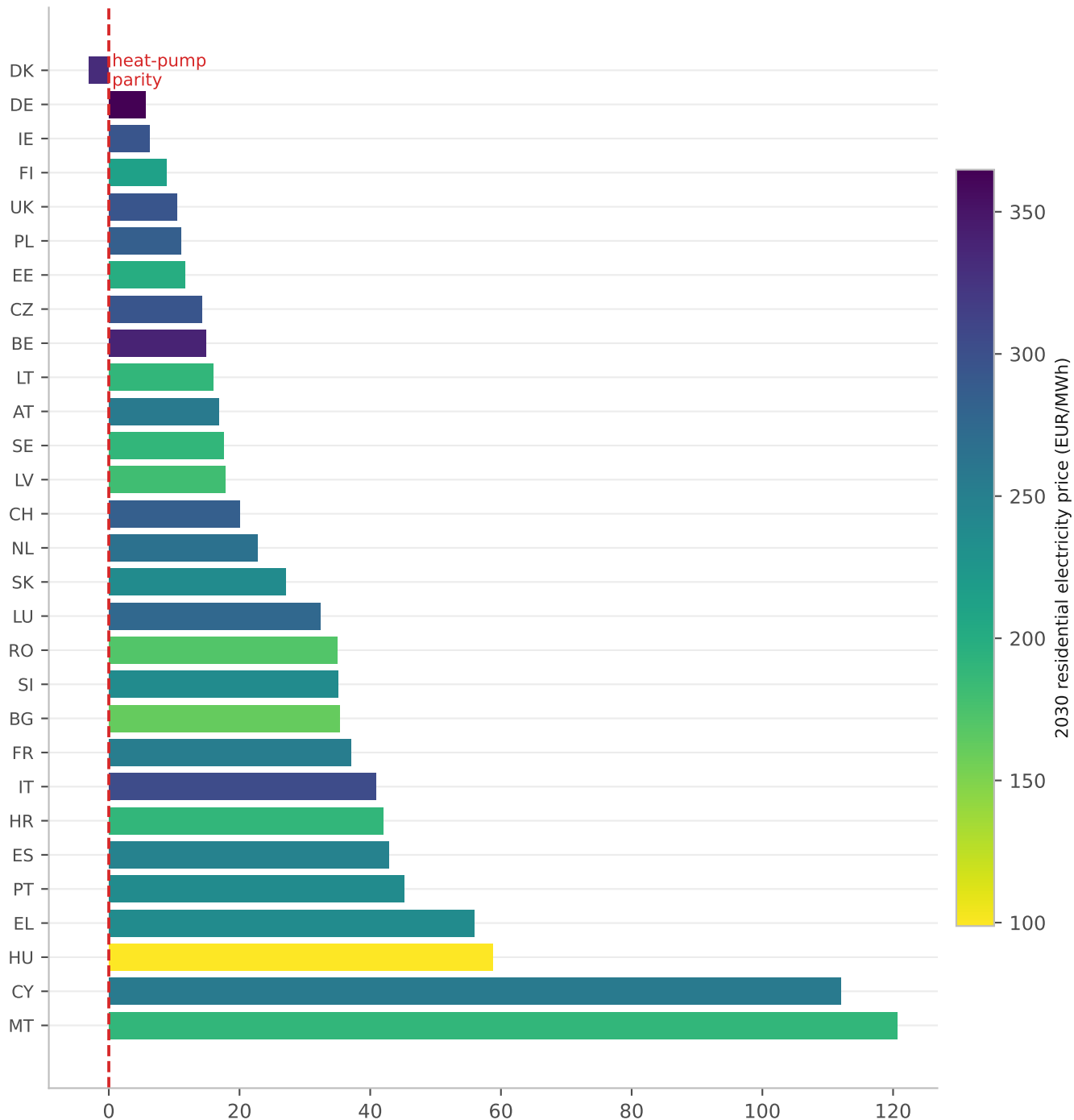


Hydrogen-boiler minus heat-pump cost by country (2050)



CENTRAL H2 trajectory. Only Germany reaches parity by 2050, driven by Europe's highest residential electricity price and high gas-grid coverage. EU mean gap ~30 EUR/MWh.

H2 boiler minus best heat pump, 2050 LCOH (EUR/MWh_user)